VELOOP Rewards

Data Analyst Internship, Task 01: Executive Summary

Data authenticity: VELOOP is a real product under development. The dataset supplied for this assignment is fictional; it was profiled and audited (Deliverables 12 and 13) and covers the Users entity only. Counts taken from it describe that file, not VELOOP. Nothing in this document is an actual VELOOP statistic. Every event, field, KPI and rule is **Proposed** or **To Be Verified**; nothing about VELOOP's product is Confirmed.

1. What was investigated

The brief asked for the foundation VELOOP needs before it can analyse real activity: what to collect, how to structure it, how to trust it, what management should measure, what engineering should implement, and what a future analytics team should see. Because VELOOP has little production data yet, the work designs that structure rather than analysing results. The starting points were the task brief and, later, the supplied simulated dataset. An attempt to find official public documentation for VELOOP Rewards found no matching official page, so no product fact has been verified. The dataset was profiled and audited; it covers the Users entity only.
The investigation covered 20 product areas (registration, tasks, earning features, daily engagement, referrals, wallet, withdrawals, support and third-party providers) and followed the brief's work packages from the data audit through to the backlog.

2. What was discovered

- **Four distinctions must be built into the data**, because mixing them would overstate results: an activity occurring is not a reward being credited; a completion claim is not a validated completion; registration is not referral conversion; and a user opening the app is not a user being active.
- **The wallet should be an append-only ledger** with a unique transaction ID and idempotent writes. Reversals and adjustments become new linked entries; originals are never edited. Without this, deduplication and reconciliation of rewards, wallet and withdrawals cannot be guaranteed.
- **Amounts must always be kept per unit.** The brief lists several reward units (VEs, Gems, Tokens, XP, Spins); they cannot be added together until a conversion rule exists.
- **Quality should be a pipeline stage, not a clean-up.** Bad records should be stopped at the earliest sensible layer and quarantined with a rule ID, never deleted or silently corrected.
- **Third parties are not under VELOOP's control.** Ads, surveys, offers and tracking can be late, wrong or silent, so a missing signal is 'unconfirmed' before it is 'failed'.
- **A practical start is enough.** A two-phase pipeline reuses what VELOOP already runs; extra infrastructure is added only when a named trigger is met.
- **The supplied dataset covers users only.** Of the 54 KPIs, 2 can be calculated in full and 5 partly; 47 need data that was not supplied.
- **The 'clean' sheet is not rule-free.** The audit logged 25 issues, including negative derived days, sessions without a login and marketing opt-in without consent; every problem injected into the audit sheet was traced to exact rows.

3. What was produced

Area	Output
Data audit and coverage	20 product areas mapped; a 42-row coverage matrix; 15 analytics gaps
Specifications	User, earning and reward, wallet and withdrawal, referral and engagement analytics
Event tracking plan	42 events (9 Critical, 16 High), one naming standard (object_action, past tense)
Data dictionary and model	128 fields (99 unique names) across 10 entities; 16 foreign keys; 20 documented relationships; ER diagram
KPI Catalogue	54 KPIs with reproducible formulas and no targets; 12 marked To Be Verified
Data quality framework	39 rules (11 Critical) across six layers, covering all 14 issue types in the brief
Dashboards	7 dashboards, 56 visuals with wireframes; every KPI appears on at least one
Engineering handoff	25 event requirements with 120 payload rows and acceptance tests; 12 general requirements; 12 pre-conditions
Pipeline, governance, change	A 13-component pipeline with an architecture diagram; an access model for 6 roles and 10 zones; a change process and log
Dataset work	Profiling, relationship tests and KPI test (2 calculable, 5 partly, 47 not); audit with 25 issues; 18 of 39 rules tested
Unknowns, gaps, backlog	50 unknowns; 31 readiness gaps; 43 backlog items

4. What is currently unknown

- **Almost everything about VELOOP's actual implementation.** 45 of 50 unknowns are To Be Verified and 3 are Proposed awaiting a decision; 2 are Confirmed, both about the supplied file. Critical unknowns: reward units (A002), a stable user ID (A003), a unique transaction ID (A005), the idempotency design (A023) and whether the wallet is append-only (A029).
- **The other datasets.** The supplied file covers the Users entity only, so relationship tests beyond Users, most KPI tests and 21 of 39 quality rules could not be run (A046).
- **Readiness.** 7 of 10 assessed areas are at level R1 (designed on paper) and 3 at R2 (checked against the supplied Users data). None has been verified with VELOOP or implemented.

5. What VELOOP should implement

Step	What	Why first
1	Settle identifiers and definitions: stable user ID, unique transaction ID, reward units, UTC timestamps, what 'active' means, what 'activated' means	Everything else joins or counts on them
2	Agree idempotent writes and the common event payload; choose the collection route and data store	Prevents duplicates and gaps before events flow
3	Instrument the 9 Critical events, then the 16 High events	Core KPIs for users, tasks, rewards, wallet and withdrawals
4	Stand up the raw zone, quarantine, transformation and the quality rules	So bad records never reach a KPI
5	Build the first dashboards (Executive Overview, Withdrawals, Wallet), then the first governed report	Reporting only after the layers beneath are trusted
6	Add Phase 2 infrastructure only where a scale trigger is met	Avoid cost that does not pay for itself

6. Major risks

Risk	Why it matters	Addressed by
Unverified foundations	The design rests on 50 unknowns; only 2 are confirmed (both about the supplied file) and 5 are Critical risk	Assumption Register; BL-30
Wallet integrity	Duplicate credits, or balances edited in place, would make wallet and withdrawal figures untrustworthy (A005, A023, A029)	BL-02, BL-04, BL-03
KPIs that change meaning	'Active user' and 'activation' are undecided, so core KPIs could shift once decided (A024, A010)	BL-07, BL-08
Mixed units	Adding amounts in different reward units would give false totals (A002)	BL-09; rule DQ035
Events for features that do not exist	Wasted effort and empty dashboards (A008)	BL-11
Simulated data mistaken for results	Fictional data used as if it were VELOOP's would mislead (brief rules 2 and 12)	Rule DQ039; BL-20
Privacy and legal unknowns	Retention, deletion and consent depend on where users are (A016, A036)	BL-29
Over-engineering	Infrastructure bought before it is needed adds cost and failure points	Two-phase pipeline with scale triggers
Rules and KPIs untested beyond Users	Only the Users data was supplied, so 21 rules and most KPIs could not be tested (A046)	BL-12

7. Major priorities

The backlog has 43 items; 6 are Critical and 17 have no dependencies and can start now. The Critical items are:

ID	Requirement	Owner
BL-01	Confirm which of the 42 proposed events already exist, who emits each, and whether each comes from the frontend or backend	Engineering
BL-02	Guarantee a stable, immutable user_id and a unique transaction_id on every wallet entry	Engineering
BL-03	Confirm the wallet ledger design: append-only or edited in place, when withdrawal amounts are held, fees and limits, cancellation, and redemption and exchange records	Engineering / Product
BL-04	Agree and test an idempotency design for event and wallet writes	Engineering
BL-05	Implement the common payload and general engineering requirements (AG-01 to AG-12)	Engineering
BL-06	Instrument the 9 Critical events to the developer handoff specification (AE-001 to AE-009)	Engineering

Decisions Product can take now: BL-03 Confirm the wallet ledger design: append-only or edited in place, when; BL-07 Decide which events make a user 'active' and the activity window; BL-08 Agree one activation definition for users and referrals, stored with a; BL-09 Define reward units and any conversion between them.

What is needed to finish: engineering confirmation of the Critical unknowns; the remaining datasets (A046), so profiling, the KPI test and the audit can be re-run; and a review of this document set with Product and Engineering, after which statuses can move from Proposed to Confirmed.